

MNYMO Competition 2024 Guts Solutions

MN YOUTH MATH OUTREACH

August 2024

1. [5] What is the absolute (positive) difference between the mean and mode of the numbers 16, 32, 21, 16, and 20?

Proposed by: Austin Wang

Answer: $\boxed{5}$

Solution: Since there are 5 numbers, the mean is $\frac{16+32+21+16+20}{5} = 21$, and the mode is 16, so the absolute difference is $21 - 16 = \boxed{5}$.

2. [5] **Sneha Kundu** and **Angie Huang** are moving around a circular track $\frac{1}{4}$ of a mile long. Because Sneha despises running, she walks at 2 miles per hour, and Angie runs at 12 miles per hour. How many laps around the track will Angie have completed once Sneha has walked one mile?

Proposed by: Austin Wang

Answer: $\boxed{24}$

Solution: At 2 miles per hour, it takes Sneha $\frac{1}{2}$ of an hour to walk one mile. In the same time frame, Angie can run $12 \cdot \frac{1}{2} = 6$ miles. Since the track is $\frac{1}{4}$ of a mile long, 6 miles is equivalent to $\boxed{24}$ laps around the track.

3. [5] Let $ABCDEF$ be a regular hexagon with side length 8. Find the area of quadrilateral $ABDF$.

Proposed by: Ryan Niu

Answer: $\boxed{64\sqrt{3}}$

Solution: Note that triangles ABD and AFD are congruent, and are $30 - 60 - 90$ triangles so $BD = FD = 8\sqrt{3}$. Thus, the area is simply $2 \cdot (\frac{1}{2} \cdot 8 \cdot 8\sqrt{3}) = \boxed{64\sqrt{3}}$.

4. [6] **Aarya Garimella** is selling slime. He has two products: The *Slime Apprentice* kit, containing 14 pieces of slime and a cleanup kit, and the *Slime Master* kit, which contains 34 pieces of slime and a cleanup kit. If the Slime Apprentice kit costs 52 dollars to produce and the Slime Master kit costs 112 dollars to produce, what is the production cost of a single piece of slime, in dollars?

Proposed by: Michael Luo

Answer: $\boxed{3}$

Solution: Let x and y be the costs of one piece of slime and a cleanup kit, respectively. We have the system

$$\begin{aligned} 14x + y &= 52 \\ 34x + y &= 112. \end{aligned}$$

Subtracting the first equation from the second gives $20x = 60$, or $x = \boxed{3}$.

5. [6] Consider a circle inscribed in an ellipse with major axis of length 6 and minor axis of length 4. Compute the area of the region outside the circle but inside the ellipse.

Proposed by: Austin Wang

Answer: $\boxed{2\pi}$

Solution: We can calculate the region as the area of the circle subtracted from the area of the ellipse. Since the circle is inscribed in the ellipse, it has a diameter of 4 (the minor axis of the ellipse). Therefore, the radius of the circle is 2, so the area of the circle is 4π . The formula for the area of an ellipse is $ab\pi$, where a is half of the major axis and b is half of the minor axis. In this problem, we have $a = 3$ and $b = 2$. Thus, the area of the ellipse is 6π . Thus, the desired area is $6\pi - 4\pi = \boxed{2\pi}$.

6. [6] Positive integers a and b satisfy $ab = 18$. Compute the smallest possible value of $a^2 + b^2$.

Proposed by: Michael Luo

Answer: $\boxed{45}$

Solution: Observe that

$$a^2 + b^2 = (a + b)^2 - 2ab = (a + b)^2 - 36,$$

therefore, it's equivalent to minimize $a + b$. This occurs when a and b are as close together as possible, or $\{a, b\} = \{3, 6\}$, which achieves $\boxed{45}$.

7. [7] The graph of $f(x) = x^2 - ax + b$ is symmetric about the line $x = 2$. Find a .

Proposed by: Austin Wang

Answer: $\boxed{4}$

Solution: Notice that the line of symmetry of a quadratic $kx^2 + lx + m$ is always $-\frac{l}{2k}$. Thus, $\frac{a}{2} = 2$, so $a = 4$.

8. [7] Aaron is going for a run. He starts from his house and runs 6 miles east, 15 miles north, 14 miles west, then runs back home in a straight line. How many miles did he run in total?

Proposed by: Austin Wang

Answer: $\boxed{52}$

Solution: Note that Aaron's total displacement from his house right before he goes home is 8 miles west and 15 miles north. Therefore, if he runs back home in a straight line, he runs for $\sqrt{8^2 + 15^2} = 17$ miles. Thus, in total, he runs for $6 + 15 + 14 + 17 = \boxed{52}$ miles.

9. [7] Right triangle $\triangle ABC$ is inscribed in a circle with diameter $BC = 5$. Let D be the foot of the altitude from A , and let AD intersect the circle at $E \neq A$. Given that $AB = 3$, find $5AE$.

Proposed by: Angie Huang

Answer: $\boxed{24}$

Solution: We have $AC = \sqrt{5^2 - 3^2} = 4$, so $\triangle ABC$ is a $3-4-5$ right triangle. Notice that $\triangle ABC$ and $\triangle DBA$ are similar, so $BD = \frac{9}{5}$ and $AD = \frac{12}{5}$. By Power of a Point, $AD \cdot DE = BD \cdot CD$, so $DE = \frac{12}{5}$. Thus $5AE = 5 \cdot \frac{24}{5} = \boxed{24}$.

10. [9] For two-digit numbers with non-zero digits, a flipped version of a number is the number when its digits' order is reversed. For example, the flipped version of 12 is 21. If the sum of a number and its flipped version is divisible by some number n for all two-digit numbers with non-zero digits, find n .

Proposed by: Ryan Niu

Answer: $\boxed{11}$

Solution: Say our number is made up of the digits a and b . Then we can represent our number as $10a + b$. Likewise, the flipped version of this number can be written as $10b + a$. When added together, we get $11a + 11b = 11(a + b)$. Thus, regardless of what the number is, the sum is always divisible by $\boxed{11}$.

11. [9] Compute $\left\lfloor \sqrt{20 - 2\sqrt{51}} \right\rfloor$. Here, $\lfloor x \rfloor$ denotes the largest integer not exceeding x .

Proposed by: Ryan Niu

Answer: $\boxed{2}$

Solution: Observe that $20 - 2\sqrt{51} = 17 + 3 - 2\sqrt{17}\sqrt{3} = (\sqrt{17} - \sqrt{3})^2$. Thus, our quantity is simply $\lfloor \sqrt{17} - \sqrt{3} \rfloor$. 17 is slightly greater than 16 and 3 is slightly less than 4, so by some simple approximations, our answer is $\boxed{2}$.

12. [9] **Sneha Kundu** is running at her school's Fun Run (We cannot say which school as it would spoil a future problem). Unfortunately, she despises running and takes a break after each lap with probability $1/2$. Suppose Sneha's gym teacher has a $1/4$ chance of seeing her each time she takes a break. Compute the expected number of laps Sneha will run before getting caught.

Proposed by: Sneha Kundu

Answer: $\boxed{8}$

Solution: Observe each lap she runs, she gets caught with probability $1/2 \cdot 1/4 = 1/8$, so the expected number of laps she runs is $\boxed{8}$.

13. [11] A 5 foot tall horse has one end of a 5 foot long rope attached to the top of its head. The other end of the rope is attached to the top of an 8 foot tall pole of negligible thickness. If the horse is always upright, compute the area of the region where the horse can roam.

Proposed by: Austin Wang

Answer: $\boxed{16\pi}$

Solution: The top of the pole is $8 - 5 = 3$ feet above the head of the horse. Let the top of the horse's head be point A , let the top of the pole be point B . Drop the altitude from point A to the pole, and let this altitude hit the pole at point C . When the rope is pulled taut, we see that the triangle ABC is a 3-4-5 right triangle. Thus, the region is a circle of radius 4, the area of which is $\boxed{16\pi}$.

14. [11] Daren and Aaron live 16 miles apart. They want to meet up somewhere in their city, however, their moms insist that they cannot travel too far. As a result, Aaron and Daren's combined displacements (distances) from their respective houses must be less than 34 miles. Compute the area of the possible points where they could meet up in the city, in square miles.

Proposed by: Ryan Niu

Answer: $\boxed{255\pi}$

Solution: Note that the description of the bounded area is simply an ellipse, where Daren and Aaron are at the two foci of an ellipse with one semi-axis of length 17 miles, and an 8 mile length from the center of the ellipse to one of the foci. Thus,

$$17^2 - 8^2 = 15^2$$

tells us that the length of the other semi-axis is 15 miles, so our area is

$$17 \cdot 15 \cdot \pi = \boxed{255\pi}.$$

15. [11] **Angie Huang, Austin Wang, Michael Luo, and Ryan Niu** are standing in front of a line of 9 rooms. An assignment of each person to a room is called *good* if each room has at most one person using it, and no two people are using adjacent rooms. Compute the number of good assignments.

Proposed by: Michael Luo

Answer: $\boxed{240}$

Solution: WLOG let Eric take the leftmost room, Austin take the second leftmost room, Jeff take the second rightmost room, and Michael take the rightmost room. Let w be the number of rooms between Jeff and the left wall plus the number of rooms between Michael and the right wall. Let x , y , and z be the number of rooms between Eric and Austin, Austin and Jeff, and Jeff and Michael, respectively.

Now the answer is the number of solutions in the integers of $w + x + y + z = 5$, where $x, y, z \geq 1$ and $w \geq 0$. This is the number of solutions in the nonnegative integers of $w + x + y + z = 2$, of which there are 10. Remembering the WLOG at the beginning, the answer is $10 \cdot 4! = \boxed{240}$.

16. [12] **Austin Wang** is making instant ramen with tofu. To make the instant ramen, Austin must perform 5 steps in the following order: add soup base, add spicy oil, add tofu, pour water, and microwave. However, Austin threw away the label, so he performed these steps in random order. Austin's microwave explodes if and only if at least one of the following conditions occur:

- The tofu is microwaved without adding water before microwaving.
- Water is poured before tofu is added and then subsequently microwaved.

Compute the probability Austin blows up his microwave.

Proposed by: Ryan Niu

Answer: $\boxed{\frac{1}{3}}$

Solution: We have two set “structures” for the possible ways to blow up the microwave. In both cases, we simply have to find the number of ways to add the other two steps (add soup base and add spicy oil) into our predetermined “structures”. If we imagine our set steps are all identical sticks, and the other two steps are a rectangle and square (distinct shapes), then we create a bijection using a pseudo-stars and bars in which we want to order five elements with three of them being distinct. Thus, for each case there are $\frac{5!}{3!} = \frac{120}{6} = 20$ ways. Since there are two disjoint cases, we have $2 \cdot 20 = 40$ total ways. The total number of ways is simply $5! = 120$, so our final probability is $\frac{40}{120} = \boxed{\frac{1}{3}}$.

17. [12] Let F_n denote the Fibonacci sequence, where $F_1 = 1$, $F_2 = 1$, and for $n > 2$, $F_n = F_{n-1} + F_{n-2}$. Compute

$$\sum_{n=1}^{\infty} \frac{F_n}{13^n}.$$

Proposed by: Ryan Niu

Answer: $\boxed{\frac{13}{155}}$

Solution: Let our desired sum be $S = \frac{1}{13} + \frac{1}{13^2} + \frac{2}{13^3} + \frac{3}{13^4} + \dots$. Then observe that

$$\frac{1}{13}S = \frac{1}{13^2} + \frac{1}{13^3} + \frac{2}{13^4} + \frac{3}{13^5} + \dots$$

If we subtract this from S , we have

$$\frac{12}{13}S = \frac{1}{13} + \frac{1}{13^3} + \frac{1}{13^4} + \frac{2}{13^5} + \dots$$

so

$$\frac{12}{13}S = \frac{1}{13} + \frac{1}{13^2}S.$$

Thus, solving for S we obtain $S = \boxed{\frac{13}{155}}$.

18. [12] Let a and b be real numbers satisfying

$$a^4 = a^2 \cdot b^2 + b^4.$$

Find the value of $(\frac{a}{b})^2$.

Proposed by: Ryan Niu

Answer: $\boxed{\frac{1 + \sqrt{5}}{2}}$

Solution: Firstly, we can divide the whole equation by $a^2 \cdot b^2$. This gives us

$$\frac{a^2}{b^2} = 1 + \frac{b^2}{a^2}.$$

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Now, we can let $\frac{a^2}{b^2} = y$ and our equation is simply

$$y = 1 + \frac{1}{y}.$$

After multiplying the whole equation by y and solving the quadratic using the quadratic equation we get $y = \frac{1 \pm \sqrt{5}}{2}$. However, note that we are using real values a and b , so $\frac{a^2}{b^2} = y$ implies y cannot be

negative. Thus, the solution $y = \frac{1 - \sqrt{5}}{2}$ is ruled out and our final answer should be $y = \boxed{\frac{1 + \sqrt{5}}{2}}$.

19. [14] The sum of the reciprocals of the factors of 2024, $(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2024})$, can be written as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.

Proposed by: Ryan Niu

Answer: $\boxed{793}$

Solution: First we start by setting the common denominator 2024 among the fractions. Observe that through this process we actually obtain the sum

$$\frac{2024}{2024} + \frac{1012}{2024} + \frac{506}{2024} + \dots + \frac{1}{2024}.$$

This means our numerator is simply the sum of all the factors of 2024 with a denominator of 2024! Thus we can finish the problem out by finding the sum of all the factors of $2024 = 2^3 \cdot 11 \cdot 23$, which is

$$(2^3 + 2^2 + 2^1 + 2^0)(11^1 + 11^0)(23^1 + 23^0) = (15)(12)(24).$$

Lastly, we can take this over 2024 and simplify to obtain the fraction $\frac{540}{253}$, so our final answer is $540 + 253 = \boxed{793}$.

20. [14] The MN Youth Math Outreach board is doing a Secret Santa. **Michael Luo** and **Austin Wang** accidentally show each other the present they bought before the exchange. If there are 7 people on the board and each person brings a unique gift, how many ways are there to distribute the gifts so that no one receives a gift they've seen?

Proposed by: Sneha Kundu

Answer: $\boxed{1280}$

Solution: Let $!n$ the number of permutations of $\{1, \dots, n\}$ with no fixed points. We claim the answer is $!7 - 2!6 - !5$. This is by complementary counting:

- There are $!7$ ways to arrange a gift exchange without restrictions.
- There are $!6$ ways to arrange a gift exchange if Michael gives a gift to Austin but Austin doesn't give a gift to Michael, by treating Michael and Austin as one unit.
- There are $!5$ ways to arrange a gift exchange if Michael gives a gift to Austin and Austin gives a gift to Michael.

Now we compute this. It's well-known (by, say, PIE) that

$$!n = n! \sum_{i=0}^n \frac{(-1)^i}{i!},$$

so we compute $!7 = 1854$, $!6 = 265$, $!5 = 44$, so the answer is $\boxed{1280}$.

21. [14] A cone made of blue Play-Doh is covered with red Play-Doh such that the outside of the red Play-Doh forms a sphere that circumscribes the blue cone. This structure is cut into two equal hemispheres with a knife such that it passes through exactly one point on the perimeter of the base of the cone. Once cut, the flat surface on both hemispheres contains red and blue areas. If the base radius of the cone is 6, and its height is $6\sqrt{3}$, find the area of the red portion.

Proposed by: Ryan Niu

Answer: $\boxed{(48 - 9\sqrt{6})\pi}$

Solution: The blue region will take shape of an ellipse, while the surrounding red region is the difference between the circular cross-section and the ellipse. Thus, it suffices to find the major and minor axes of the ellipse to find its area. Since the base radius of the cone is 6, and the height of the cone is $6\sqrt{3}$, the cone represents an equilateral triangle from a side view perspective. Thus, the major axis of the ellipse is simply $6\sqrt{3}$. Still considering this equilateral triangle perspective, denote the top point as A and the other two points as B and C . We see the center of the ellipse lies on the midpoint of the major axis, so using 30-60-90 triangles, our semi-major axis is $3\sqrt{3}$, so the height of the center from the base of triangle ABC is $\frac{3\sqrt{3}}{2}$. If we draw line segment DE with D on AB and E on AC , with DE parallel to BC , then we can see by similar triangles ADE and ABC , the side length of ADE is 9. Now, considering the 3-D model again, we know that our minor axis lies on the circle of radius 9, and is a distance $\frac{3}{2}$ from the center, so by the Pythagorean Theorem, our semi-minor axis is $3\sqrt{2}$. This gives us that the blue region has area $3\sqrt{3} \cdot 3\sqrt{2} \cdot \pi = 9\sqrt{6} \cdot \pi$. The red region is then just the area of the circle of the same radius as the sphere, which is 48π . Thus, our final area is $\boxed{(48 - 9\sqrt{6})\pi}$.

22. [16] Consider a triangle ABC with lengths $AB = 13$, $BC = 14$, and $AC = 15$. Consider the circles with diameter AB and AC . Let these circles intersect at point P other than A . Let $\overline{AP}$ intersect the common external tangent of the circles at point X such that X and A are on opposite sides of $\overline{BC}$. Compute the length of AX .

Proposed by: Austin Wang

Answer: $\boxed{6 + 4\sqrt{3}}$

Solution: Recall that a $13 - 14 - 15$ triangle may be decomposed into a $5 - 12 - 13$ right triangle and a $9 - 12 - 15$ right triangle. Thus, since the circles are centered at the midpoints of AB and BC , they both pass through the foot of the altitude from B to AC . We are given that this point is P , so $AP = 12$.

Now, since AP is the common chord of the two circles, it is also the radical axis of the circles. Thus, X lies on the radical axis of the circles. Let the tangent points be B_1 and C_1 , respectively. Since X lies on the radical axis and on $\overline{B_1C_1}$, we know that X is the midpoint of $\overline{B_1C_1}$.

Denote the centers of the circles as O_b and O_c , respectively. Drop the altitude from O_b to $\overline{O_cC_1}$. Let the foot of the altitude be H . We can see that $\overline{O_bO_c} = \frac{1}{2}\overline{BC} = 7$, and $\overline{O_cH} = \frac{15}{2} - \frac{13}{2} = 1$. Then by the Pythagorean Theorem, we see $\overline{O_bH} = \sqrt{49 - 1} = 4\sqrt{3}$. Thus, we see that $\overline{XB_1} = 2\sqrt{3}$. Let the length of AX be x . From power of a point, we see that $x(x - 12) = 2\sqrt{3}^2$. Rearranging, we get $x^2 - 12x - 12 = 0$. Solving this quadratic, we find that $x = 6 \pm 4\sqrt{3}$. Since lengths can never be negative, we conclude that $x = \boxed{6 + 4\sqrt{3}}$.

23. [16] Let $a_1, \dots, a_n$ be a finite geometric sequence of positive integers with $n > 2$. Given that a_1 and a_n both have exactly 6 divisors and a_2 has exactly k divisors, find the sum of all possible values of k .

Proposed by: Sam Kretzschmar

Answer: $\boxed{24}$

Solution: We note that a positive integer N can only have exactly 6 factors iff it is of the form $N = p^5$ or $N = p^2q$ for primes p, q . Thus we have three cases if $a_1 \neq a_n$:

1. Say $a_1 = p^5$ and $a_n = q^5$ for primes $p, q \neq p$. As 5 is prime, the only finite geometric series possible with the two with $n > 2$ is $p^5, p^4q, p^3q^2, p^2q^3, pq^4, q^5$. This means $a_2 = p^4q$ and $k = (1 + 1)(4 + 1) = 10$.

2. Say $a_1 = p^5$ and $a_n = q \cdot r^2$ for primes p, q, r , with $q \neq r$. To have this be possible with the lone q , we set $p = q$, which yields p^5, p^3r, pr^2 , so $k = 8$.

3. Say $a_1 = p^2q$ and $a_n = r^2s$ for primes p, q, r, s with $p \neq q$ and $r \neq s$. To avoid $n = 2$ being forced, we set $q = s$. Then the sequence p^2q, pqr, r^2q works for this, with $a_2 = pqr$ so $k = 2^3 = 8$.

The final possibility is that $a_1 = a_n$, which would yield $k = 6$ as well. These add to $10 + 8 + 6 = 24$.

24. [16] Find the last three digits of the expression $1^{100} + 3^{100} + 5^{100} + \dots + 19^{100}$.

Proposed by: Austin Wang

Answer: 258

Solution: This question is asking for the residue of this expression modulo 1000. We will take this expression modulo 8 and modulo 125. Using Euler's theorem, we see that $1^4 \equiv 3^4 \equiv \dots \equiv 19^4 \equiv 1 \pmod{8}$, so $1^{100} + 3^{100} + \dots + 19^{100} \equiv 10 \equiv 2 \pmod{8}$. Similarly, in mod 125, all such i^{100} are congruent to 1 mod 125 except 5^{100} and 15^{100} , which are clearly congruent to 0 mod 125. Thus, $1^{100} + 3^{100} + \dots + 19^{100}$ is congruent to 8 mod 125. Therefore, our expression is congruent to 2 mod 8 and 8 mod 125, so by the Chinese remainder theorem it is congruent to 258 modulo 1000.

25. [20] Let N be the number of primes $p < 10^6$ such that $p + 2$ is also prime. Estimate N .

An estimate of $E > 0$ earns $\lfloor 20 \min(N/E, E/N) \rfloor$ points.

Proposed by: Michael Luo

Answer: 8169

Solution: From the prime number theorem, the “probability” n is prime is approximately $\frac{1}{\log n}$, therefore, the probability n and $n + 2$ are both prime is approximately

$$\frac{1}{\log n} \cdot \frac{1}{\log(n+2)} \approx \frac{1}{\log^2 n}.$$

Therefore, we may approximate the answer as

$$\sum_{n=2}^{10^6} \frac{1}{\log^2 n}.$$

Estimating $\log^2 n \approx \log^2 10^6$ and $\log 10 \approx 2.3$ gives the sum as

$$\frac{10^6}{(6 \cdot 2.3)^2} \approx 5250,$$

which is good enough for 8 points.

With some care, we can get better estimates. Observe that if n is a prime, n is odd, so $n + 2$ is odd too, multiplying the probability $n + 2$ is prime by 2. Now for any odd prime q , the probability $p + 2$ isn't divisible by q is $\frac{q-2}{q-1}$ since $p \not\equiv -2 \pmod{q}$. Therefore, the fact that p is prime multiplies the probability $p + 2$ is prime by

$$\frac{\frac{q-2}{q-1}}{\frac{q-1}{q}} = 1 - \frac{1}{(q-1)^2}.$$

Taking the product of these influences over all primes, we get

$$2 \prod_{q \text{ prime}, q \neq 2} \left(1 - \frac{1}{(q-1)^2} \right) \approx 1.32.$$

A closer estimate of the sum

$$\sum_{n=2}^{10^6} \frac{1}{\log^2 n} \approx \int_2^{10^6} \frac{1}{\log^2 n} \approx 6247$$

gives the answer $6247 \cdot 1.32 \approx 8246$, which is good for 19 points.

The exact answer may be computed using the following Python code:

```

1 def isprime(n):
2     for i in range(2, int(n ** 0.5) + 1):
3         if n % i == 0:
4             return False
5     return True
6
7 count = 0
8
9 for n in range(2, 1000000):

```

```

10     if isprime(n) & isprime(n + 2):
11         count += 1
12
13 print(count)

```

26. [20] The MN Youth Math Outreach board consists of **Angie Huang, Aarya Garimella, Jinu Lee, Sneha Kundu, Ryan Niu, Michael Luo, and Austin Wang**, who are all from distinct high schools. Estimate the area A of the convex hull formed by these schools in square miles.

An estimate $E > 0$ earns $\lfloor 20 \min(A/E, E/A) \rfloor$ points.

(The *convex hull* of a finite set of points $\mathcal{P}$ is the smallest convex polygon which contains all points in $\mathcal{P}$. It may be visualized as a rubber band stretched around $\mathcal{P}$.)

Proposed by: Michael Luo

Answer: 2126.96

Solution: Recall that Jinu goes to Moorhead, Michael goes to Minnetonka, Austin goes to Mounds View, and everyone else goes to school in the general Twin Cities area. We make the following approximations:

- The convex hull is the triangle formed by Moorhead, Minnetonka, and Mounds View. Furthermore, this triangle has a right angle at Minnetonka.
- It takes 3 hours to drive from Minnetonka to Moorhead at 60 mph.
- It takes 30 minutes to drive from Minnetonka to Mounds View at 30 mph.

From this, the area is approximately

$$\frac{1}{2} \cdot 180 \cdot 15 = 1350$$

square miles, which would earn 12 points.

We may compute the exact distance using Google Maps. Angie, Aarya, Jinu, Sneha, Ryan, Michael, and Austin go to East Ridge, Edina, Moorhead, Wayzata, Woodbury, Minnetonka, and Mounds View, respectively.

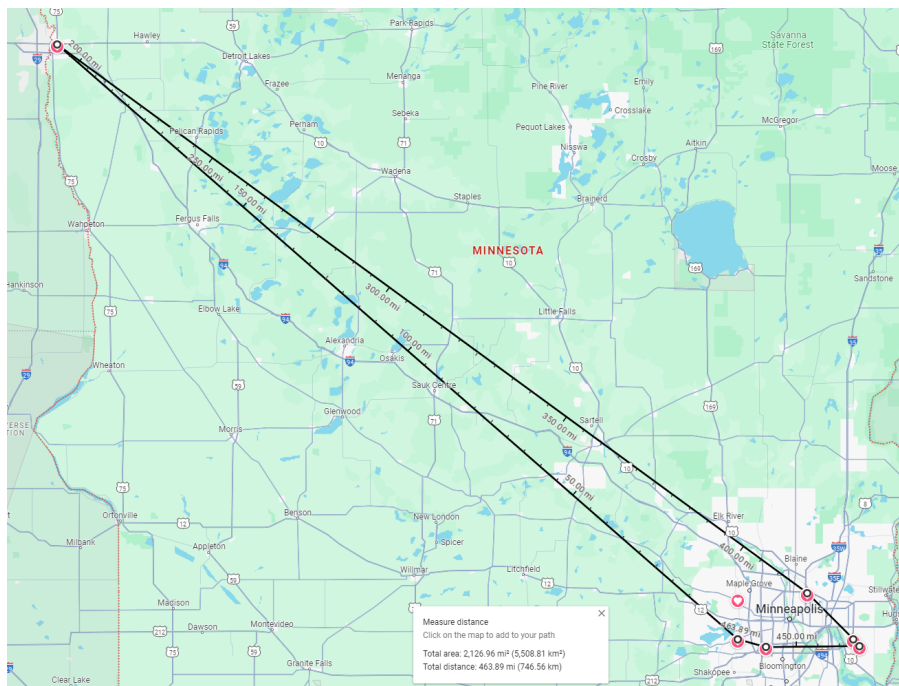


Figure 1: The schools in question with their convex hull. The Minnesota state border is highlighted. Note that most of the schools are in the Twin Cities area, and Wayzata is the only school not on the border.

The Google Maps tool gives the area as 2126.96 square miles.

27. [20] The MN Youth Math Outreach Board has birthdays $B_1, B_2, \dots, B_7$ when written in MMD-DYYYY form. For example, if Taylor Swift was the eighth member of the MN Youth Math Outreach Board, B_8 would be 12131989, since her birthday is December 13th, 1989. Estimate the sum S of the pairwise gcds of the B_i , or

$$S = \sum_{1 \leq i < j \leq 7} \gcd(B_i, B_j).$$

An estimate of E earns $\lfloor 20 \min(S/E, E/S) \rfloor$ points.

Proposed by: Michael Luo

Answer: 390

Solution: Assume the birthdays are randomly chosen integers from 1 to 12000000. Let $N = 12000000$, we compute the expected gcd of two integers chosen randomly from 1 to N . It's well-known that the probability of two integers chosen from $\{1, 2, \dots, N\}$ being relatively prime is approximately $\frac{6}{\pi^2}$, so the probability of the gcd being k for some $k \leq N$ is $\frac{6}{(k\pi)^2}$. Therefore we estimate the expected value

$$\sum_{k \leq N} \frac{6}{k\pi^2} = \frac{6}{\pi^2} \sum_{k \leq N} \frac{1}{N} \approx \frac{6}{\pi^2} \log N \approx 10.$$

Since there are $\binom{7}{2}$ possible pairs, our estimate is 210, which earns 10 points.

Now we compute the exact answer. The birthdays are listed in the table, and the answer 390 may be computed with the Python code.

Name	Angie	Aarya	Jinu	Sneha	Ryan	Michael	Austin
Birthday	02122008	02212007	11092006	07012008	05272008	09282008	12172008

```

1 from math import gcd
2
3 birthdays = [2122008, 2212007, 11092006, 7012008, 5272008, 9282008, 12172008]
4
5 total = 0
6
7 for i in birthdays:
8     for j in birthdays:
9         total += gcd(i, j)
10
11 print(0.5 * (total - sum(birthdays))) # correct for overlap

```